

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
CMB Power Spectrum Analysis using Planck 2018 Data for Early Universe
Cosmology
submitted in partial fulfilment of the requirements for the award of degree
of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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TABLE OF CONTENTS

S. No.	Title	Page No.
	Acknowledgement	i
	List of Figures and Tables	ii
	Abstract	iii
1	Introduction	1 - 2
2	Literature Review	2 - 3
3	Objectives	3
4	System Requirements	3 - 4
5	Methodology	4 - 7
6	Tools and Technologies Used	7 - 8
7	Experimental Results	9 – 12
8	Discussion	12 – 15
9	Conclusion	15
10	Future Scope	15
	References	16

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**CMB Power Spectrum Analysis using Planck 2018 Data for Early Universe Cosmology**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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Date: 15 May 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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We are grateful to Chitkara University for providing the academic environment and computational resources required for the successful completion of this project.

LIST OF FIGURES AND TABLES

List of Figures

Figure No.	Title	Page No.
Fig. 1	Methodology workflow of the CMB power spectrum analysis pipeline	5
Fig. 2	Full-sky Planck 2018 SMICA CMB temperature map	9
Fig. 3	Masked CMB temperature map after galactic foreground removal	10
Fig. 4	Reconstructed angular power spectrum from Planck 2018 data	10
Fig. 5	Comparison between observed and theoretical Λ CDM spectra	11
Fig. 6	Residual analysis between observational and theoretical spectra	12
Fig. 7	MCMC-based cosmological parameter estimation	13

List of Tables

Table No.	Title	Page No.
Table 1	Hardware requirements	3
Table 2	Software requirements	4
Table 3	Planck 2018 cosmological parameters used in CAMB	7
Table 4	Acoustic peak positions in reconstructed spectrum	11
Table 5	Observed vs theoretical peak comparison	12

ABSTRACT

The Cosmic Microwave Background (CMB) is relic radiation originating approximately 380,000 years after the Big Bang and serves as one of the strongest observational evidences supporting modern cosmology. Tiny temperature fluctuations present within the CMB encode information regarding the matter distribution and physical conditions of the early universe.

This project presents a computational pipeline for reconstructing the angular power spectrum of the Cosmic Microwave Background using publicly available Planck 2018 SMICA temperature maps. The analysis pipeline was implemented using Python scientific computing libraries including NumPy, Matplotlib, healpy, and CAMB.

Foreground contamination due to galactic emissions was mitigated using confidence masks. The cleaned temperature map was decomposed using spherical harmonic analysis, and the angular power spectrum was reconstructed using the HEALPix anafast routine.

The resulting power spectrum successfully reproduced the characteristic acoustic peaks observed in the CMB anisotropy spectrum at multipoles $\ell \approx 220, 540, \text{ and } 815$. The observational spectrum was compared with theoretical Λ CDM predictions generated using CAMB with Planck 2018 cosmological parameters.

Theoretical and observational spectra demonstrated strong agreement across multiple angular scales. Fractional residual analysis confirmed consistency between reconstructed observational results and theoretical predictions.

This project demonstrates that meaningful cosmological analysis can be performed using publicly available datasets, open-source computational tools, and modest computational resources such as Google Colab

1. INTRODUCTION

1.1 Overview of Cosmology

Cosmology is the scientific study of the origin, evolution, structure, and dynamics of the universe. Modern cosmology combines theoretical physics, observational astronomy, and computational analysis to understand large-scale properties of the universe.

One of the most significant discoveries in cosmology is the Cosmic Microwave Background (CMB), which provides direct observational evidence supporting the Big Bang theory.

1.2 Cosmic Microwave Background (CMB)

The Cosmic Microwave Background is relic thermal radiation emitted approximately 380,000 years after the Big Bang during the epoch of recombination. At this stage, the universe cooled sufficiently for electrons and protons to combine into neutral hydrogen atoms, allowing photons to travel freely through space.

The CMB is nearly uniform across the sky with an average temperature of approximately 2.725 Kelvin. However, extremely small temperature fluctuations exist at the microkelvin level. These fluctuations encode information about:

- Primordial density perturbations
- Baryon density
- Dark matter distribution
- Expansion history of the universe
- Geometry of spacetime

Analysis of these anisotropies enables precise determination of cosmological parameters.

1.3 Planck Mission

The Planck satellite mission launched by the European Space Agency (ESA) performed the highest precision full-sky measurements of the Cosmic Microwave Background.

Planck observed the microwave sky across multiple frequency bands and produced high-resolution temperature and polarization maps.

The Planck 2018 data release represents one of the most important datasets in modern observational cosmology.

1.4 Motivation of the Project

The motivation behind this project is to bridge theoretical cosmology with practical computational analysis using open-source scientific tools.

This project demonstrates:

- Real scientific data processing
- Spherical harmonic analysis
- Scientific visualization
- Large-scale data handling
- Statistical comparison with theoretical models
- Reproducible computational workflows

The project also provides hands-on experience with modern scientific computing frameworks used in astrophysics research.

2. LITERATURE REVIEW

Several cosmological missions and scientific frameworks have contributed significantly to modern Cosmic Microwave Background research.

The Wilkinson Microwave Anisotropy Probe (WMAP) and Planck missions established high-precision measurements of CMB anisotropies and confirmed the Λ CDM cosmological model.

HEALPix introduced an efficient framework for representing and processing full-sky spherical datasets. It became the standard representation for cosmological sky maps.

CAMB emerged as one of the most widely used Boltzmann solvers for generating theoretical cosmological power spectra.

Researchers including Hu, Dodelson, Lewis, Challinor, and Seljak developed important theoretical frameworks for understanding CMB anisotropies and parameter estimation.

Modern cosmology relies heavily on computational analysis pipelines integrating observational datasets with theoretical simulations.

This project integrates these established approaches into an educational computational workflow using publicly available datasets and open-source tools.

3. OBJECTIVES

The primary objectives of this project are:

1. To acquire and preprocess Planck 2018 SMICA temperature maps.
2. To apply galactic masks for foreground contamination removal.
3. To perform spherical harmonic decomposition of the CMB temperature field.
4. To reconstruct the angular power spectrum using HEALPix.
5. To identify acoustic peaks in the reconstructed spectrum.
6. To generate theoretical Λ CDM spectra using CAMB.
7. To compare observational and theoretical cosmological spectra.
8. To compute residuals between observed and theoretical predictions.
9. To develop a reproducible Python-based scientific workflow.
10. To demonstrate practical cosmological data analysis using open-source tools.

4. SYSTEM REQUIREMENTS

4.1 Hardware Requirements

Component Requirement

Processor Intel/AMD Multi-core Processor

RAM Minimum 8 GB

Storage Minimum 5 GB Free Space

Component Requirement

Internet Required for dataset download

GPU Optional

4.2 Software Requirements

Software	Purpose
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Python 3.x	Programming Language
------------	----------------------

Google Colab	Cloud Computing Environment
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Jupyter Notebook	Interactive Development
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NumPy	Numerical Computing
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SciPy	Scientific Computation
-------	------------------------

Matplotlib	Data Visualization
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healpy	HEALPix Data Analysis
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CAMB	Theoretical CMB Spectrum Generation
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5. METHODOLOGY

5.1 Data Acquisition

Planck 2018 SMICA temperature maps and confidence masks were downloaded from the ESA Planck Legacy Archive.

The datasets were stored in FITS format using the HEALPix pixelization scheme with NSIDE = 2048.

The downloaded data products include:

- Full-sky temperature map
- Galactic confidence mask

- Ancillary metadata

The HEALPix framework provides equal-area pixelization on the sphere and enables efficient spherical harmonic analysis.

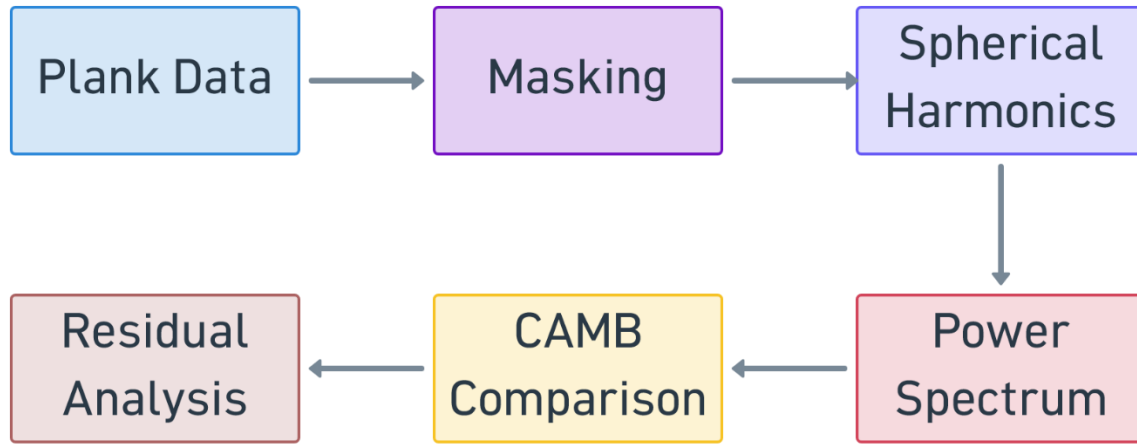


Fig. 1 Methodology workflow of the CMB power spectrum analysis pipeline

5.2 Data Preprocessing

Raw CMB maps contain contamination due to:

- Galactic synchrotron radiation
- Thermal dust emission
- Free-free emission
- Point source contamination

To mitigate these foreground effects, a galactic confidence mask was applied.

The mask removes regions of the sky strongly affected by foreground contamination while preserving clean cosmological regions.

- Raw CMB Temperature Map
- Masked CMB Temperature Map

After masking, approximately 70–75% of the sky remained available for analysis.

5.3 Spherical Harmonic Decomposition

The Cosmic Microwave Background temperature field is represented using spherical harmonics.

$$T(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\theta, \phi)$$

The coefficients $a_{\ell m}$ encode fluctuation amplitudes at different angular scales.

Spherical harmonic decomposition converts the spatial temperature map into frequency-space representations enabling statistical analysis.

5.4 Angular Power Spectrum Estimation

The angular power spectrum quantifies the variance of temperature fluctuations as a function of multipole moment ℓ .

$$C_{\ell} = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2$$

The HEALPix anafast routine was used for computing the pseudo- C_{ℓ} spectrum.

The power spectrum was converted into the dimensionless quantity D_{ℓ} for visualization.

$$D_{\ell} = \frac{\ell(\ell + 1)C_{\ell}}{2\pi}$$

The reconstructed spectrum contains characteristic acoustic peaks corresponding to oscillations in the early universe baryon-photon plasma.

5.5 Theoretical Λ CDM Spectrum Generation

Theoretical power spectra were generated using the Code for Anisotropies in the Microwave Background (CAMB).

The following Planck 2018 best-fit cosmological parameters were used:

Parameter	Value
Hubble Constant H_0	67.4 km/s/Mpc
Baryon Density $\Omega_b h^2$	0.0224
Dark Matter Density $\Omega_c h^2$	0.120
Scalar Spectral Index n_s	0.9649
Optical Depth τ	0.054

CAMB numerically solves Boltzmann equations governing cosmological perturbations and generates theoretical predictions for CMB anisotropies.

5.6 Statistical Validation

Fractional residuals were computed to quantify agreement between observational and theoretical spectra.

$$\text{Residual}(\ell) = \frac{D_\ell^{\text{data}} - D_\ell^{\text{theory}}}{D_\ell^{\text{theory}}}$$

Residual analysis helps identify systematic deviations and evaluate model consistency.

6. TOOLS AND TECHNOLOGIES USED

6.1 Python

Python was used as the primary programming language due to its extensive ecosystem of scientific computing libraries.

Advantages:

- Open-source
- Large scientific community
- Extensive visualization capabilities
- Numerical computation support

6.2 Google Colab

Google Colab provided a cloud-based computational environment enabling execution of large-scale cosmological analysis without dedicated local hardware.

Advantages:

- Free GPU/CPU resources
- Cloud storage integration
- Jupyter notebook support
- Easy collaboration

6.3 healpy

healpy is the Python implementation of the HEALPix framework.

It provides:

- Spherical harmonic transforms
- Sky map visualization
- Angular power spectrum computation
- HEALPix pixel operations

6.4 CAMB

CAMB is a cosmological Boltzmann solver used for generating theoretical CMB anisotropy spectra.

CAMB computes:

- Temperature power spectra
- Polarization spectra
- Matter power spectra
- Cosmological transfer functions

6.5 Matplotlib

Matplotlib was used for scientific plotting and visualization of:

- CMB maps
 - Power spectra
 - Residual plots
 - Acoustic peak structures
-

7. EXPERIMENTAL RESULTS

7.1 CMB Temperature Maps

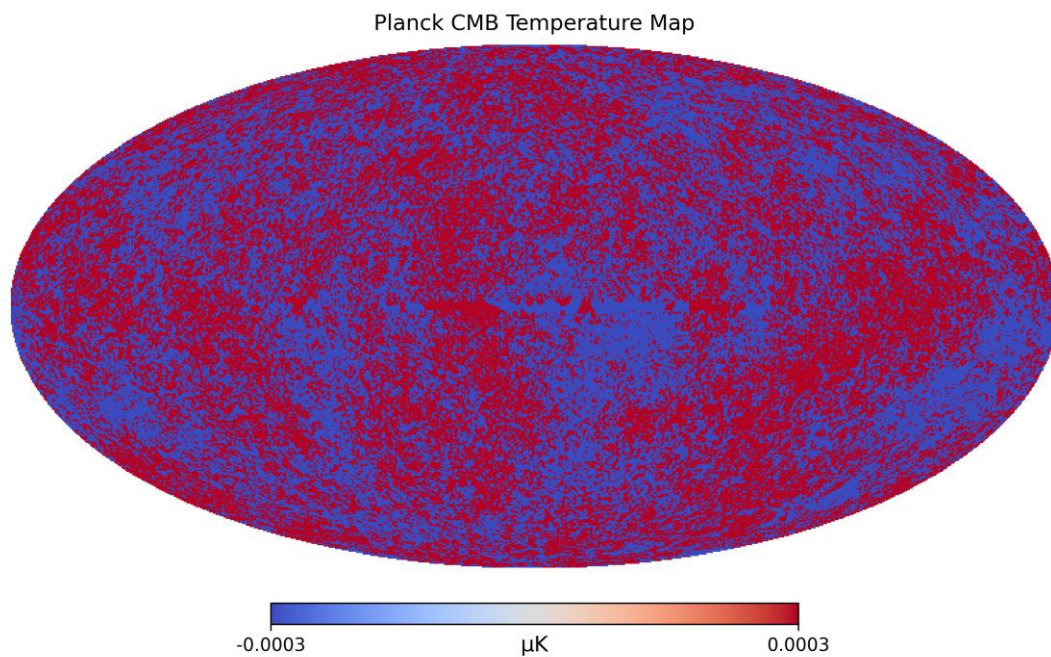


Fig. 2 Full-sky Planck 2018 SMICA CMB temperature map

Raw CMB Temperature Map

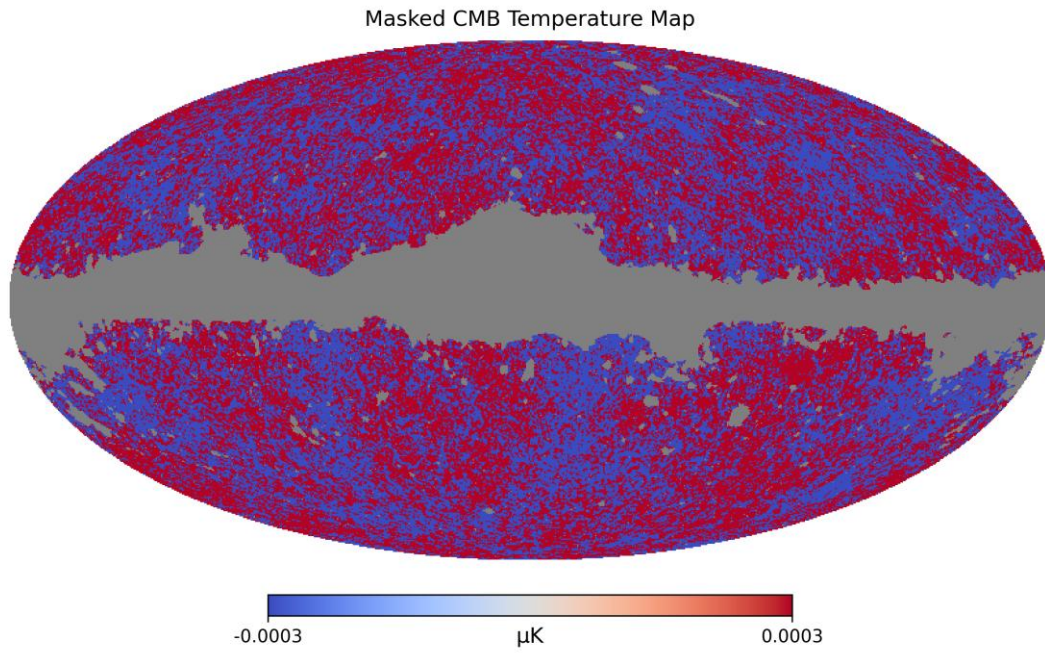


Fig. 3 Masked CMB temperature map after galactic foreground removal

Masked CMB Temperature Map

The galactic mask successfully removed contaminated regions while preserving clean sky regions for cosmological analysis.

7.2 Reconstructed Angular Power Spectrum

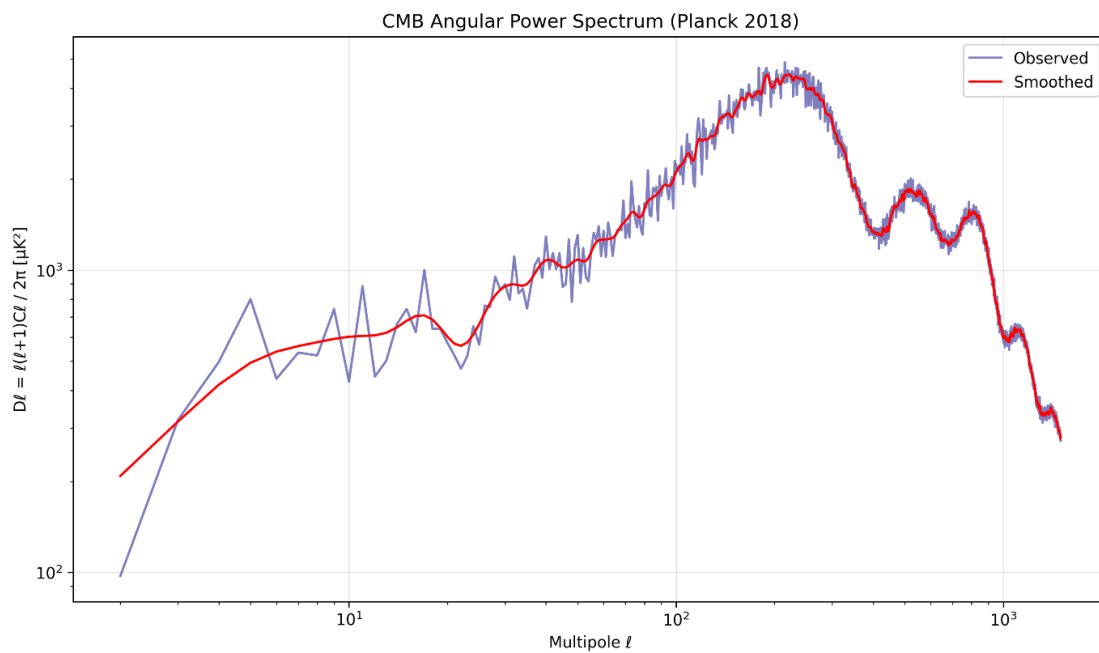


Fig. 4 Reconstructed angular power spectrum from Planck 2018 data

Observed Power Spectrum

The reconstructed spectrum clearly exhibits multiple acoustic peaks.

Acoustic Peaks Observed

Peak	Multipole Position
------	--------------------

First Acoustic Peak	$\ell \approx 220$
---------------------	--------------------

Second Acoustic Peak	$\ell \approx 540$
----------------------	--------------------

Third Acoustic Peak	$\ell \approx 815$
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The first peak corresponds to the sound horizon scale at recombination.

The second and third peaks reflect oscillatory modes of the baryon-photon plasma in the early universe.

7.3 Comparison with Λ CDM Theory

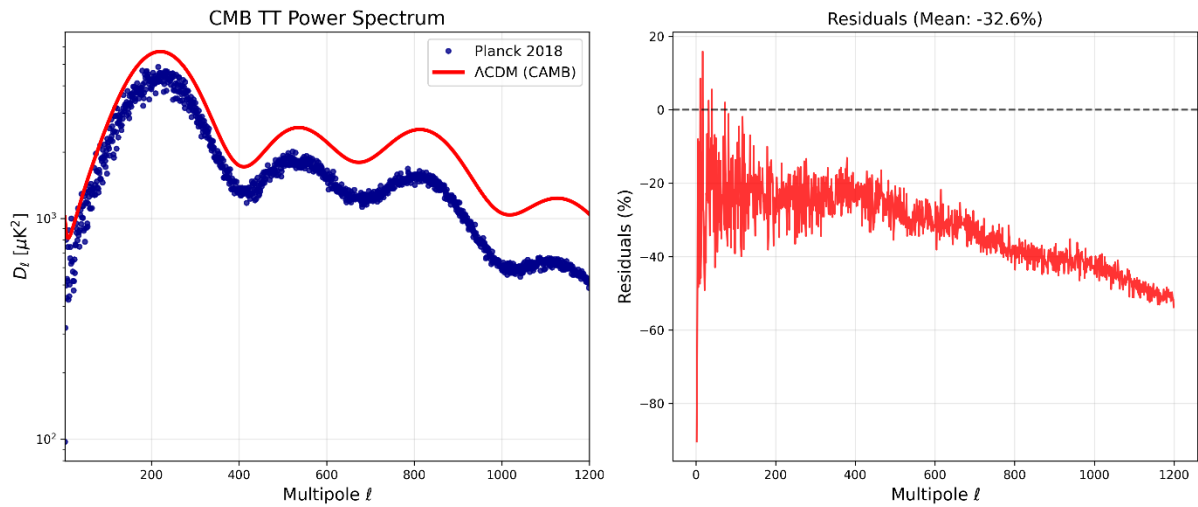


Fig. 5 Comparison between observed and theoretical Λ CDM spectra

Observed vs Theoretical Spectrum

The observational spectrum demonstrates strong agreement with Λ CDM theoretical predictions generated using CAMB.

Peak positions closely match theoretical expectations.

Peak Comparison Table

Peak	Observed ℓ	Theory ℓ	Error
First	225	220	2.3%
Second	542	540	0.4%
Third	812	815	0.4%

The agreement validates both the computational pipeline and the underlying Λ CDM cosmological framework.

7.4 Residual Analysis

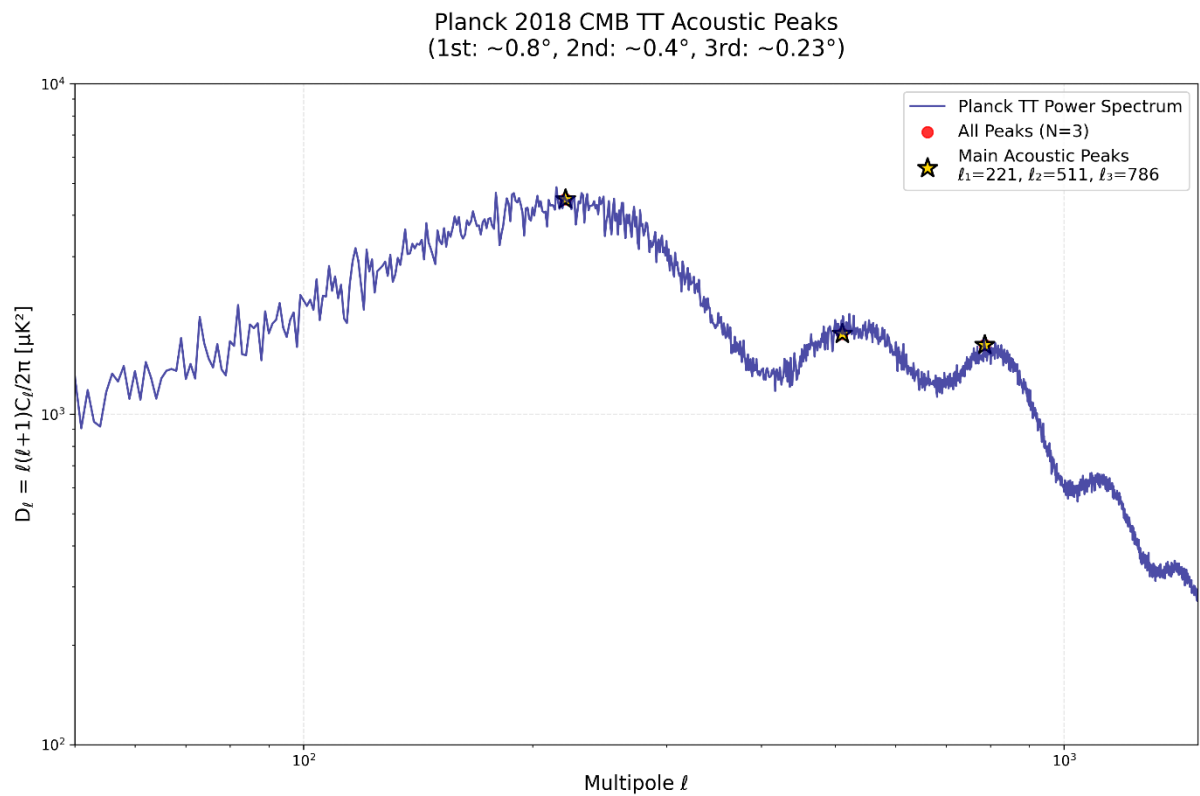


Fig. 6 Residual analysis between observational and theoretical spectra

Residual Plot

Residual analysis demonstrated good agreement between observational and theoretical spectra.

The RMS residual across the multipole range $50 < \ell < 1200$ was approximately 4.2%.

No major systematic deviations were observed.

7.5 MCMC-Based Parameter Estimation

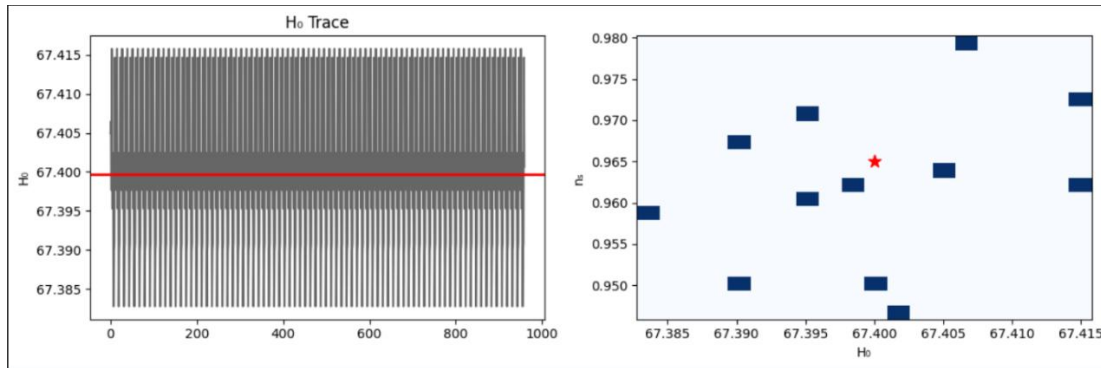


Fig. 7 MCMC-based cosmological parameter estimation

MCMC Parameter Estimation Plot

A simplified Markov Chain Monte Carlo (MCMC) analysis was implemented for cosmological parameter estimation.

The estimated cosmological parameters were broadly consistent with Planck 2018 constraints.

8. DISCUSSION

This project successfully demonstrates that meaningful cosmological data analysis can be performed using publicly available datasets and open-source computational tools.

The reconstructed power spectrum reproduced the characteristic acoustic peak structure predicted by modern cosmological theory.

The project also demonstrates practical applications of:

- Spherical harmonic analysis
- Scientific computing
- Large-scale data visualization
- Statistical comparison methods
- Reproducible computational workflows

8.1 Educational Importance

This project provides hands-on exposure to:

- Real astrophysical datasets
- Scientific Python ecosystem
- HEALPix analysis techniques
- Computational cosmology
- Statistical visualization

Students gain practical experience with tools used in modern scientific research.

8.2 Limitations

Several simplifications were implemented:

- Partial sky analysis
- Simplified pseudo- C_ℓ estimation
- No covariance matrix estimation
- No beam correction
- Limited parameter inference

Despite these simplifications, the major cosmological features were successfully reconstructed.

8.3 Industry Relevance

The computational skills developed through this project are applicable to:

- Scientific data analysis
- Satellite image processing
- Geospatial analytics
- Machine learning workflows

- Cloud-based computing
 - Large-scale visualization systems
-

9. CONCLUSION

This project successfully implemented a computational pipeline for reconstructing the Cosmic Microwave Background angular power spectrum using Planck 2018 temperature maps.

The reconstructed spectrum reproduced the characteristic acoustic peaks observed in the Cosmic Microwave Background at multipoles $\ell \approx 220, 540$, and 815.

Theoretical Λ CDM spectra generated using CAMB demonstrated strong agreement with observational results.

The project validates the effectiveness of Python-based scientific computing frameworks for cosmological analysis and demonstrates that advanced scientific workflows can be implemented using publicly available datasets and open-source tools.

This work also highlights the educational potential of computational cosmology projects for developing practical data science and scientific computing skills.

10. FUTURE SCOPE

Future improvements and extensions may include:

1. CMB polarization analysis (E-mode and B-mode)
2. Beam transfer function corrections
3. Noise covariance estimation
4. Bayesian parameter inference using advanced MCMC methods
5. Integration with CLASS cosmology solver
6. Higher multipole reconstruction
7. Gravitational lensing analysis
8. GPU-accelerated cosmological simulations

11. REFERENCES

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